DNA Methylation Analysis: Statistics and Machine Learning in Breast Cancer

Thesis Work Presentation
Master’s Degree in Mathematical Engineering

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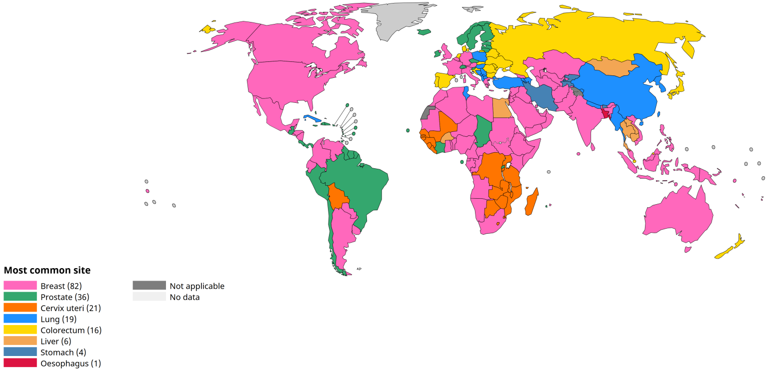
Introduction: Goals of the Presentation

- Background in **Mathematical Engineering** with a focus on **Data Analysis**.
- Goal: present a concrete thesis project integrating mathematics, computer science, and biomedicine.
- Case study: *DNA methylation* analysis in **breast cancer**.

A quantitative perspective on a real biomedical dataset.

Breast Cancer: Context and Relevance

- **High global incidence**, affecting millions of individuals.
- **Early detection** strongly improves patient outcomes.
- Why data matters: quantitative analysis supports **screening**, **diagnosis**, and **risk assessment**.

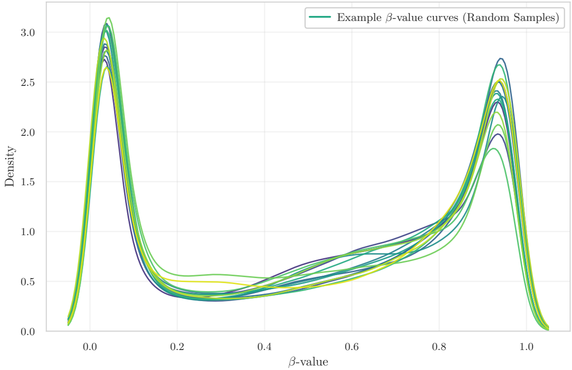


Most common cancer site per country, 2022 (excl. NMSC).

Source: Cancer TODAY — IARC, International Agency for Research on Cancer Globocan 2022.

CpG and β-values: The Language of Epigenetic Data

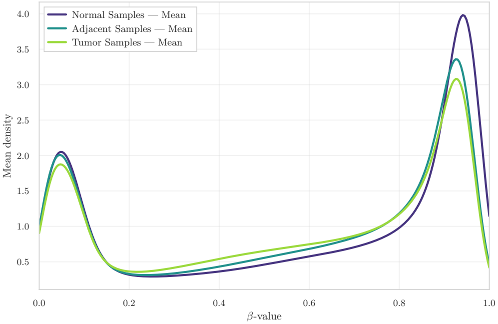
- **CpG sites**: genomic positions where DNA methylation can occur.
- β**-value** $\in [0, 1]$: proportion of methylation at each CpG.
- **DNA methylation** modulates **gene regulation**.
- In cancer, both **hypermethylation** and **hypomethylation** emerge.



Example β-value density curves from random samples. Internal dataset visualisation (thesis exploratory analysis).

Field Cancerization: The Central Research Question

- **Central question: is "normal" tissue near the tumor already epigenetically altered?**
- As Dr. Gambino highlights:
  "Identify early methylation alterations in normal tissues and assess their role in tumor initiation."
- Concept of **field cancerization**.



Mean β-value densities for Normal, Adjacent, and Tumor samples. Internal dataset visualisation (exploratory analysis).

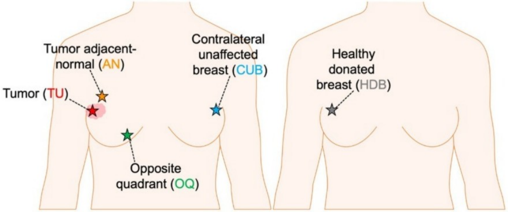


Illustration of sampling locations in breast tissue. Conceptual illustration of tumor, adjacent, and healthy breast tissue.

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Accessing and Understanding the Data

- GEO datasets: **heterogeneous formats** and **incomplete metadata**.
- Small n, huge p (up to $\sim$800k CpG sites).
- Raw **IDAT files** vs. preprocessed β**-value matrices**.
- **Unknown or inconsistent preprocessing** in some datasets.

Dataset	Samples (n)	CpGs (p)
GSE69914	397	485,512
GSE225845	477	750,426
GSE287331	311	702,573

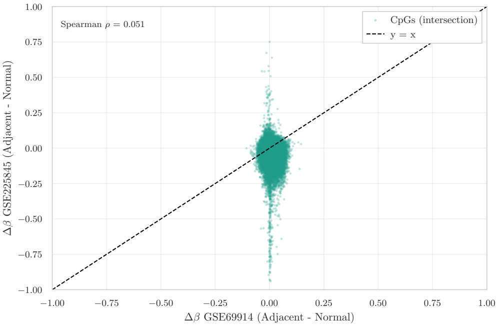
Summary of the three GEO datasets used in the analysis.

- **Huge files** $\rightarrow$ use of **Parquet** format, **LZ4** compression, chunked loading.
- Strict RAM constraints on Kaggle environments.
- Need for **Illumina manifest alignment** to correctly map probe IDs.

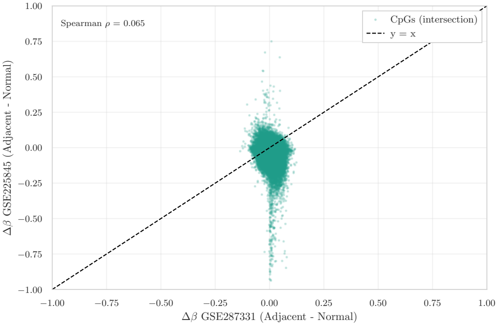
Memory-aware processing pipeline. From raw huge files to compressed Parquet storage, enabling analysis under strict RAM limits.

Can I Merge the Datasets? NO

- Different array platforms (**HM450** vs. **EPIC**).
- **Heterogeneous** and partly **unknown preprocessing pipelines**.
- Direct merge would introduce strong batch and platform effects.
- Decision: perform all analyses **within each dataset** (intra-dataset).



$\Delta\beta$ correlation between GSE69914 and GSE225845.
Intersection CpGs show minimal concordance across platforms.



$\Delta\beta$ correlation between GSE287331 and GSE225845.
Cross-dataset signal remains inconsistent, confirming strong batch effects.

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3 Exploratory Data Analysis (EDA)

Overall Research Workflow

3 Exploratory Data Analysis (EDA)

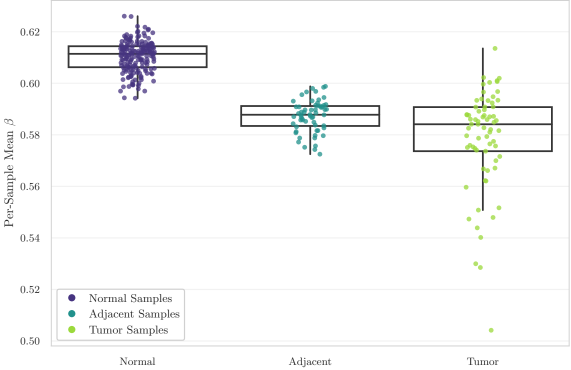
- **Consultation of medical, statistical, and computational literature** to understand biological context and methodological constraints.
- **Reproduction of key results from published studies** to confirm the consistency and reliability of each dataset.
- **Progressive methodological innovation**: integrating statistical filtering, stratification strategies, dimensionality reduction, and ultimately ML/DL modelling with careful hyperparameter tuning.

Goal: build a robust, reproducible, and biologically-consistent analysis pipeline.

Global DNA Methylation Distributions

3 Exploratory Data Analysis (EDA)

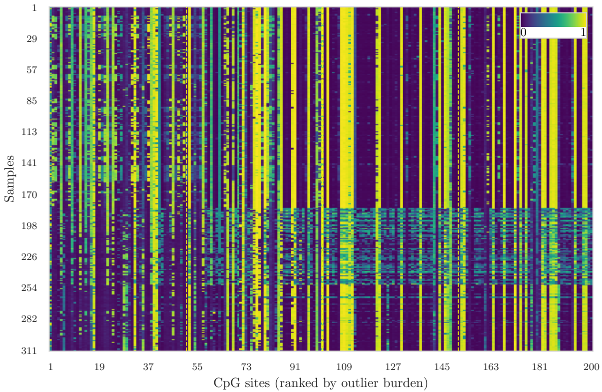
- Canonical bimodal β**-value distribution** (unmethylated vs. highly methylated CpGs).
- **Tumor samples**: shift towards **hypomethylation** and increased heterogeneity.
- **Adjacent samples**: early, subtle deviations from the Normal baseline.



Per-sample mean β-value distributions across tissue groups.
Tumor samples show global hypomethylation and higher variability compared to Normal and Adjacent samples.

Variability and Epigenetic Outliers

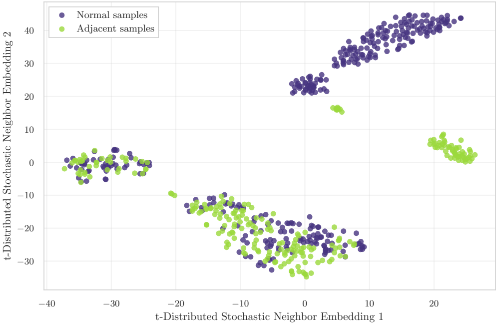
- **Unstable CpGs** are identified as those showing the highest number of extreme methylation values across samples.
- These CpGs highlight regions showing **focal hyper- or hypomethylation**.
- Local methylation instability is a hallmark of **biologically relevant epigenetic alterations**.



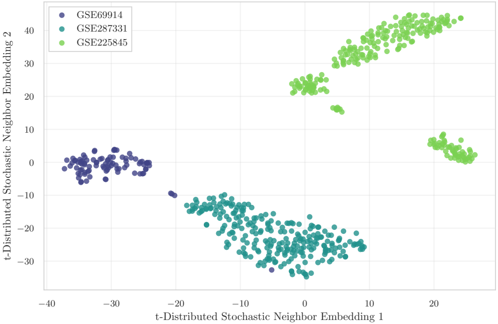
Heatmap of the 200 most unstable CpGs.
Colours represent β-values (blue = low methylation, yellow = high methylation).

Dataset Embeddings and Group Structure

- **GSE69914**: strong overlap between tissue groups.
- **GSE225845**: more complex and diffuse embedding structure.
- **GSE287331**: clear separation between **Normal** and **Adjacent**.



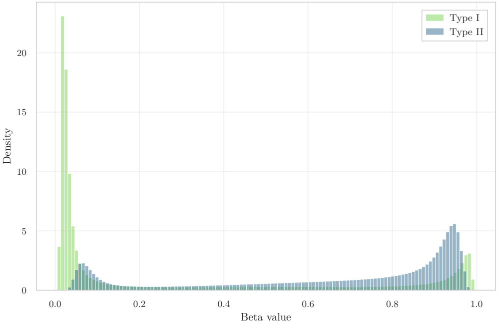
t-SNE embedding colored by tissue state.
Shows differences between Normal and Adjacent samples.



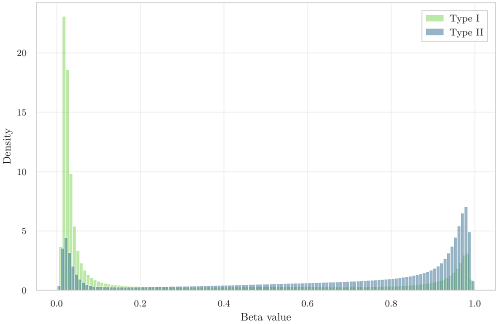
t-SNE embedding colored by dataset.
Highlights strong dataset-specific clustering and batch effects.

Probe-Design Bias and BMIQ Normalization

- Two probe designs (**Type I** vs. **Type II**) with incompatible β-value distributions.
- **GSE287331** shows a strong uncorrected probe-design bias.
- **BMIQ** performs quantile mapping of Type II probes toward the Type I distribution.
- Caution: BMIQ may reduce biological variability and partially mask real signals.



Pre-BMIQ: Type I vs. Type II distributions.
Strong mismatch between probe types highlights probe-design bias.



Post-BMIQ: Type I vs. Type II distributions.
BMIQ aligns Type II probes to Type I, but may dampen biological signal.

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Feature Selection in High Dimensions

- $p \gg n \rightarrow$ **dimensionality reduction** and filtering are essential.
- Technical filters: cross-reactive probes, SNP probes, non-CpG probes, Illumina manifest.
- Evaluate and correct potential batch effects (e.g., **COMBAT**).
- Three families of feature selection strategies:
  - **Statistical**: $\Delta\beta$, variance filtering, stability measures.
  - **Biological**: promoters, CpG islands, curated gene sets.
  - **ML-based**: Lasso, decision trees, embedded model selection.

ML in Non-Ideal Conditions

- **Train / validation / test** with extremely small n (tens of samples).
- Generalising across datasets (A $\rightarrow$ B) is often **unreliable** due to batch and platform effects.
- **How many features** to select? **Which metrics** are stable in the $p \gg n$ regime?
- **Overfitting**, noise, and limited interpretability of complex models.
- Deep Learning: training from scratch is **infeasible** $\rightarrow$ only **pre-trained / transfer learning** approaches are realistic.

Approach	Data need	Interpretability	Robustness (small n)	Typical use-case here
Statistical	low–medium	high	high	$\Delta\beta$, variance, QC summaries
ML	medium–high	medium	medium–low	classification, feature selection
DL	very high	low	low	transfer learning / pre-trained models only

Comparison of modelling approaches under small-n / large-p constraints. Summary of data requirements, interpretability, robustness, and typical use-cases in this thesis.

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Take-home Message

- **Real-world bioinformatics pipeline**: from noisy GEO data and incomplete metadata to a **clean, analysable** methylation matrix via file conversion, manifest alignment, probe filtering, and batch assessment.
- **Epigenetics is deeply quantitative**: hundreds of thousands of CpGs, $p \gg n$, non-Gaussian β-distributions, strong batch and probe-design effects.
- **Role of mathematics and statistics**: exploratory analysis, stability and variability measures, dimensionality reduction, feature selection, and careful model evaluation under small-n/large-p constraints.
- **Machine learning in non-ideal conditions**: limited data, strong regularisation, preference for simpler and interpretable models; deep learning only via **transfer learning** or as a complement, not as a starting point.

Big picture: working at the interface of **biology, statistics, and computing** is challenging but extremely rewarding — and research projects like this are much closer to our skill set than they might appear.

Thank you!

If you have any questions, I'd be happy to answer.